

IBM APPLIED DATA SCIENCE CAPSTONE

Will the First Stage Land?

Predicting SpaceX Falcon 9 landing success to price a launch

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GitHub repository: <https://github.com/kautzarichramsyah/Applied-Data-Science-Capstone>

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Executive Summary

90

Falcon 9 launches analysed
(2010–2020)

66.7%

first stages that landed
successfully

83.3%

test accuracy (15 of 18
launches)

84.4%

SVM accuracy across 50
data splits

Methods: collected launch data from the SpaceX REST API and Wikipedia, cleaned it with pandas, explored it with SQL, seaborn, Folium and a Plotly Dash app, then tuned four classifiers with GridSearchCV.

What drives landing success: experience. Success rose from 0% (2010–2013) to 84–90% (2019–2020), and newer boosters (FT, B4, B5) land far more often than v1.0/v1.1.

Where and what: KSC LC-39A has the best record (76.9%); ES-L1, GEO, HEO and SSO orbits landed every time, GTO only 51.9%.

Prediction: all four tuned classifiers reach 83.3% on the 18-launch test set; across 50 resampled splits SVM is the best and most stable model (84.4%). Errors are false positives.

Introduction

Space Y wants to bid against SpaceX — so it needs to know what a Falcon 9 launch really costs

BACKGROUND

SpaceX advertises Falcon 9 launches at about \$62 million; other providers charge upwards of \$165 million. Most of the saving comes from landing and reusing the first stage.

WHY IT MATTERS

If we can predict whether the first stage will land, we can estimate the true cost of a launch — the information a competing bidder needs.

Questions we set out to answer

- How do payload mass, orbit, launch site and flight number affect landing success?
- Has landing success improved over time, and with which booster versions?
- Can a machine learning model predict a successful landing — and how reliably?

01

Methodology

Data collection, wrangling, EDA, interactive analytics and classification

Methodology Overview

Executive summary of the end-to-end pipeline



Tools: Python (requests, BeautifulSoup, pandas, NumPy, scikit-learn, seaborn, Folium, Plotly Dash) and SQLite via ipython-sql.

Outcome variable: Class = 1 if the first stage landed successfully (ocean, ground pad or drone ship), 0 otherwise.

Code: all notebooks and spacex_dash_app.py are in <https://github.com/kautzarichramsyah/Applied-Data-Science-Capstone>

Data Collection — SpaceX REST API

1

GET request

api.spacexdata.com/
v4/launches/past



2

Normalize

response.json() →
pd.json_normalize()



3

Enrich

Look up rocket,
launchpad, payload
and core IDs



4

Filter

Keep Falcon 9 only
(drop Falcon 1) → 90
launches



5

Clean

Fill missing
PayloadMass with the
mean

Key code

```
response = requests.get(static_json_url)
data = pd.json_normalize(response.json())
getBoosterVersion(data); getLaunchSite(data); getPayloadData(data); getCoreData(data)
data_falcon9 = df[df['BoosterVersion'] != 'Falcon 1']
data_falcon9['PayloadMass'] =
data_falcon9['PayloadMass'].fillna(data_falcon9['PayloadMass'].mean())
```

Data Collection — Web Scrapping

1

Request

GET the Wikipedia
"List of Falcon 9 and
Falcon Heavy
launches" page



2

Parse

Load the HTML into a
BeautifulSoup object



3

Find tables

`soup.find_all("table")`
→ launch tables



4

Extract

Column names from
<th>, rows from <td>



5

Save

Dictionary →
DataFrame → CSV

WHY A SECOND SOURCE

Wikipedia adds human-readable mission details (customer, launch outcome, booster landing) that cross-check the API records.

CHECK

`soup.title` returns "List of Falcon 9 and Falcon Heavy launches - Wikipedia", confirming the right page was parsed before extracting rows.

Data Wrangling

Turning eight landing outcomes into one binary label

Outcome	Meaning	Count	Class
True ASDS	Landed on drone ship	41	1
None None	No landing attempted	19	0
True RTLS	Landed on ground pad	14	1
False ASDS	Drone ship landing failed	6	0
True Ocean	Controlled ocean landing	5	1
False Ocean	Ocean landing failed	2	0
None ASDS	Drone ship, no attempt	2	0
False RTLS	Ground pad landing failed	1	0

LABEL

landing_class = 0 if outcome in
bad_outcomes else 1 → df['Class']

RESULT

Mean of Class = 0.667: two out of three
Falcon 9 first stages landed. CCAFS SLC
40 hosted 55 of the 90 launches.

EDA and Interactive Visual Analytics — Methodology

Charts (seaborn)

- Scatter / strip plots: flight number, payload vs site and orbit
- Bar chart: success rate by orbit
- Line chart: yearly success trend
- One-hot encoding of categorical features for modelling

SQL (SQLite)

- Loaded SpaceX.csv into SPACEXTBL
- SELECT DISTINCT, LIKE, SUM, AVG, MIN
- Subqueries and GROUP BY / ORDER BY
- Date filtering with substr()

Folium + Dash

- Circles and markers for each launch site
- MarkerCluster of green/red outcomes
- PolyLine distances to coast and infrastructure
- Dash: site dropdown, pie chart, payload slider, scatter

Predictive Analysis — Methodology



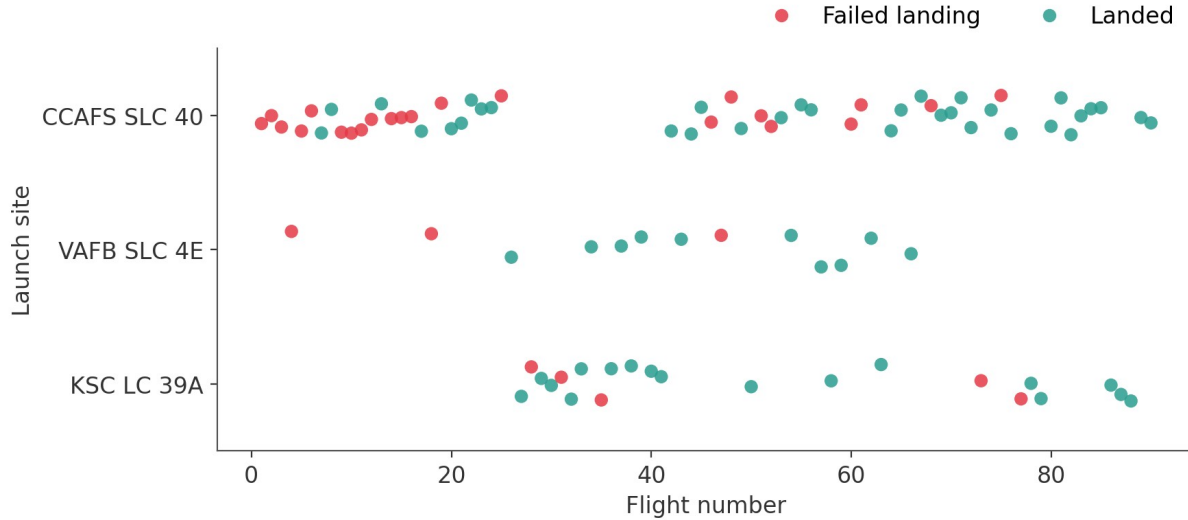
Model	Hyperparameters searched
Logistic regression	$C \in \{0.01, 0.1, 1\}$, penalty = l2, solver = lbfgs
Support vector machine	kernel $\in \{\text{linear, rbf, poly, sigmoid}\}$, C and gamma $\in \text{logspace}(-3, 3, 5)$
Decision tree	criterion, splitter, max_depth 2-18, min_samples_leaf, min_samples_split
K-nearest neighbours	n_neighbors 1-10, algorithm, p $\in \{1, 2\}$

02

EDA with Visualization

What the scatter plots, bar and line charts reveal

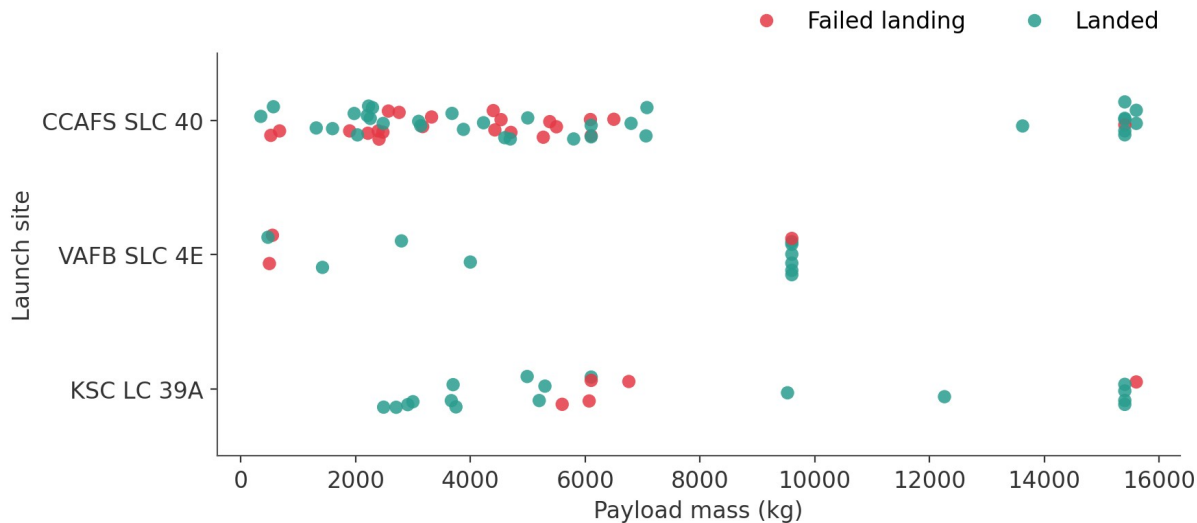
Flight Number vs. Launch Site



INSIGHT

Early flights (mostly from CCAFS SLC 40) failed to land. As flight numbers grow, green dominates at every site — experience matters more than location. KSC LC 39A only joined after flight 26 and started strong.

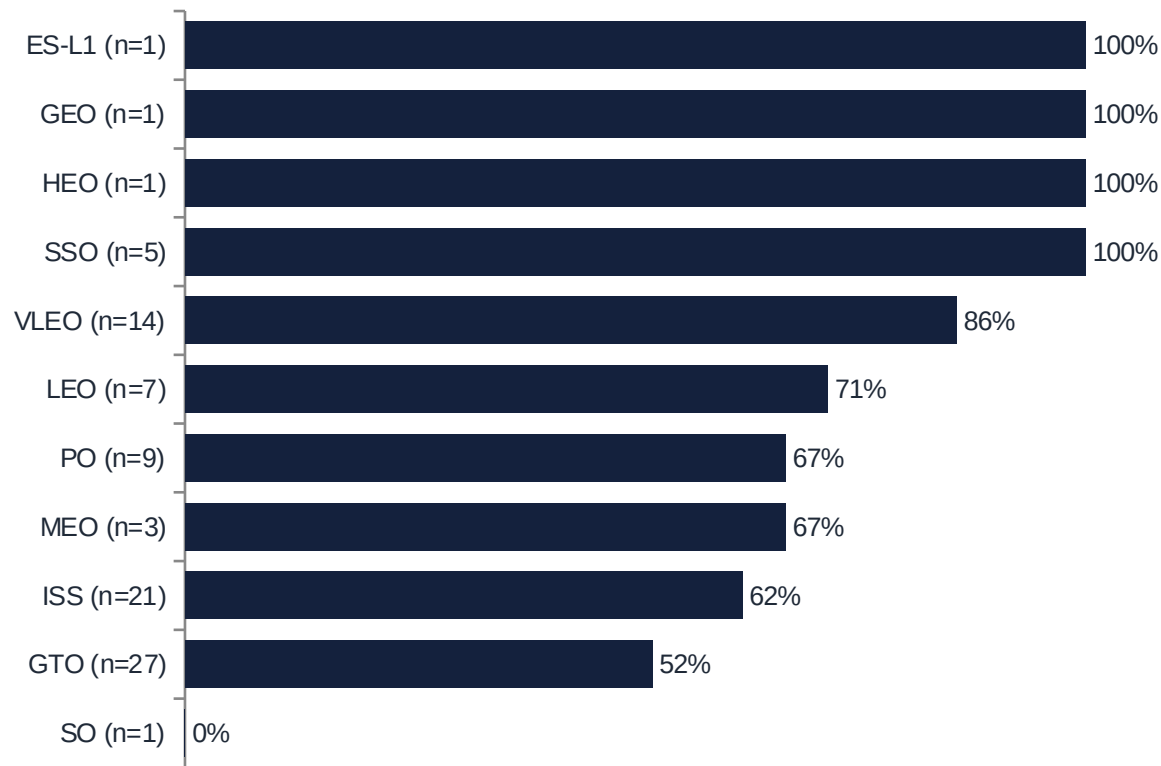
Payload Mass vs. Launch Site



INSIGHT

VAFB SLC 4E launched no payloads above 10,000 kg. Heavy payloads (> 8,000 kg) mostly landed (87%), largely because they flew late on Block 5 boosters.

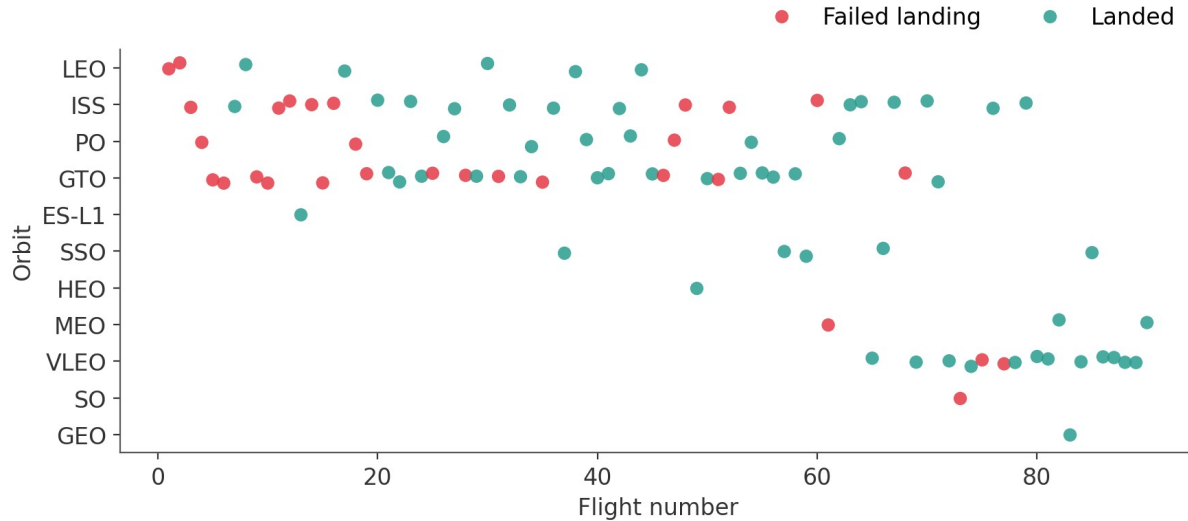
Success Rate by Orbit Type



INSIGHT

ES-L1, GEO, HEO and SSO landed every time, but with 1-5 launches each. Among the busy orbits, VLEO (86%) and LEO (71%) beat ISS (62%) and GTO (52%), the hardest orbit to recover from.

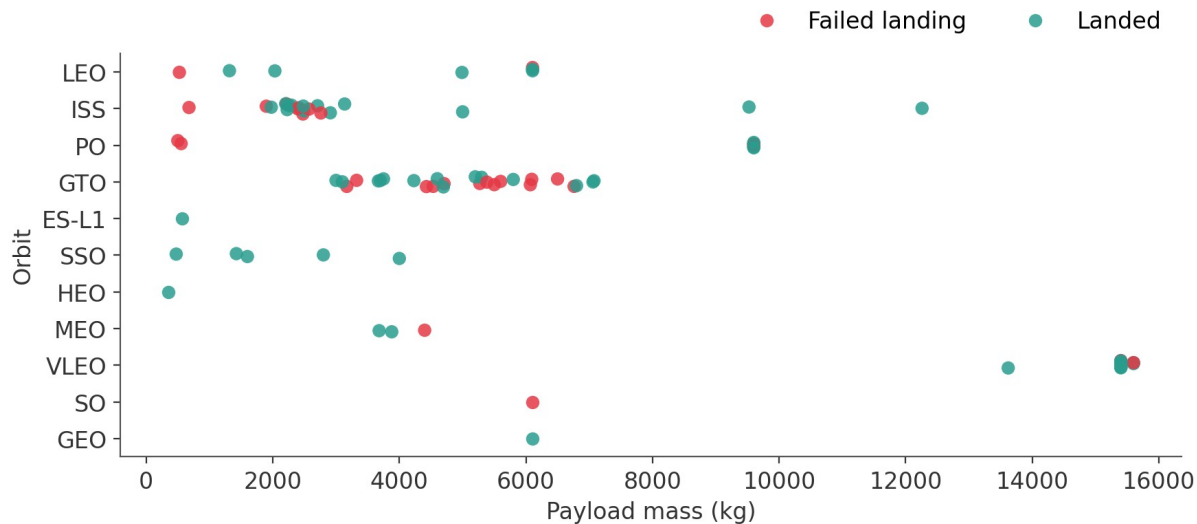
Flight Number vs. Orbit Type



INSIGHT

In LEO, success clearly tracks flight number. In GTO there is no such trend: failures continue into the later flights, so orbit energy, not just experience, limits recovery.

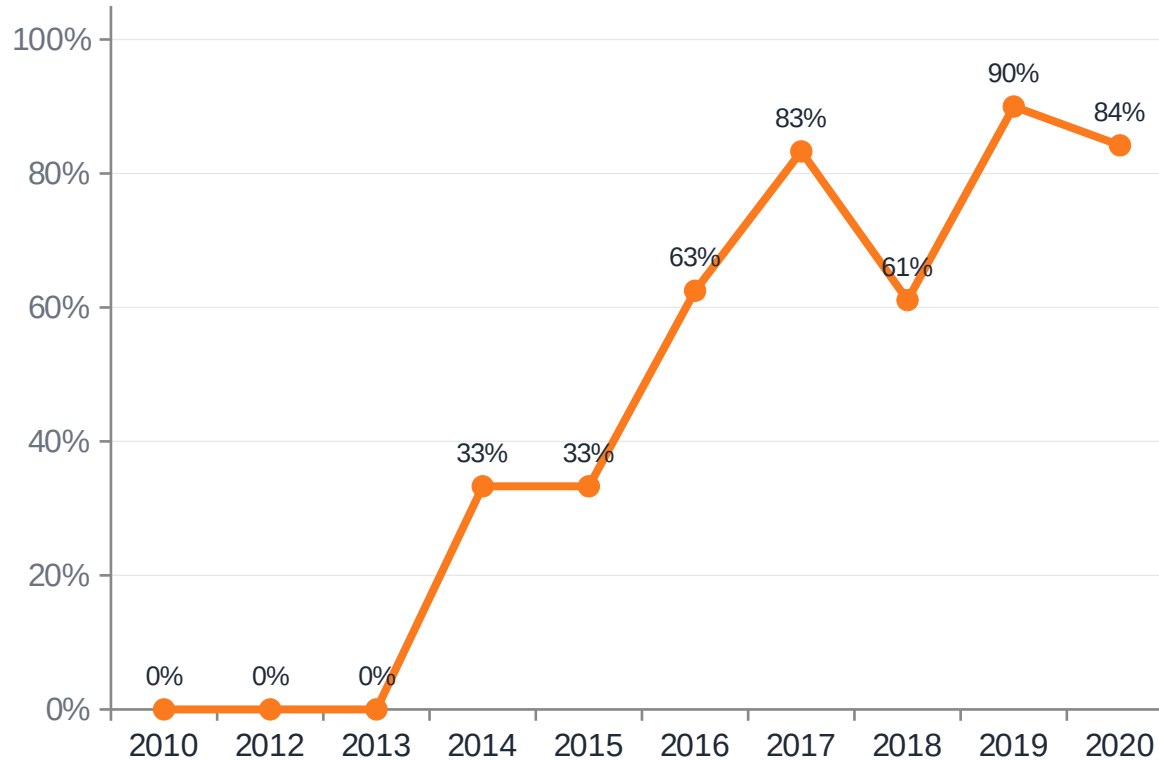
Payload Mass vs. Orbit Type



INSIGHT

Heavy payloads land well in Polar, LEO, ISS and VLEO orbits. GTO payloads (3,000–7,000 kg) are a mix of successes and failures, with no clear payload threshold.

Launch Success Yearly Trend



INSIGHT

No landings succeeded from 2010 to 2013. The rate climbed steadily from 2013, dipped in 2018 as Block 5 was introduced, and reached 90% in 2019.

EDA with SQL — Results (1/2)

SQLite on the SPACEXTBL table (Spacex.csv, 101 rows)

#	Question	Query (core)	Result
1	Unique launch sites	SELECT DISTINCT Launch_Site FROM SPACEXTBL	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
2	5 records where site begins with "CCA"	SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5	First 5 launches, all CCAFS LC-40
3	Total payload for NASA (CRS)	SELECT SUM(PAYLOAD_MASS__KG_) ... WHERE Customer = 'NASA (CRS)'	45,596 kg
4	Average payload, F9 v1.1	SELECT AVG(PAYLOAD_MASS__KG_) ... WHERE Booster_Version = 'F9 v1.1'	2,928.4 kg
5	First successful ground-pad landing	SELECT MIN(Date) ... WHERE Landing_Outcome = 'Success (ground pad)'	22 December 2015

EDA with SQL — Results (2/2)

SQLite on the SPACEXTBL table (Spacex.csv, 101 rows)

#	Question	Query (core)	Result
6	Drone-ship successes with payload 4,000–6,000 kg	<code>... WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000</code>	F9 FT B1022, B1026, B1021.2, B1031.2
7	Mission outcomes	<code>SELECT CASE WHEN Mission_Outcome LIKE 'Success%' ... GROUP BY Outcome</code>	100 success, 1 failure
8	Boosters that carried the max payload	<code>WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL)</code>	12 Block 5 boosters (e.g. B1048.4, B1049.4, B1051.3)
9	2015 drone-ship failures	<code>WHERE substr(Date,1,4) = '2015' AND Landing_Outcome = 'Failure (drone ship)'</code>	Jan: B1012, Apr: B1015 (CCAFS LC-40)
10	Landing outcomes ranked, 2010-06-04 to 2017-03-20	<code>GROUP BY Landing_Outcome ORDER BY COUNT(*) DESC</code>	No attempt 10; drone-ship success 5 and failure 5; ground pad 3 ...

03

Launch Sites Proximity Analysis

Interactive maps built with Folium

Folium Map — All Launch Sites

folium.Circle + folium.Marker at each site's latitude / longitude

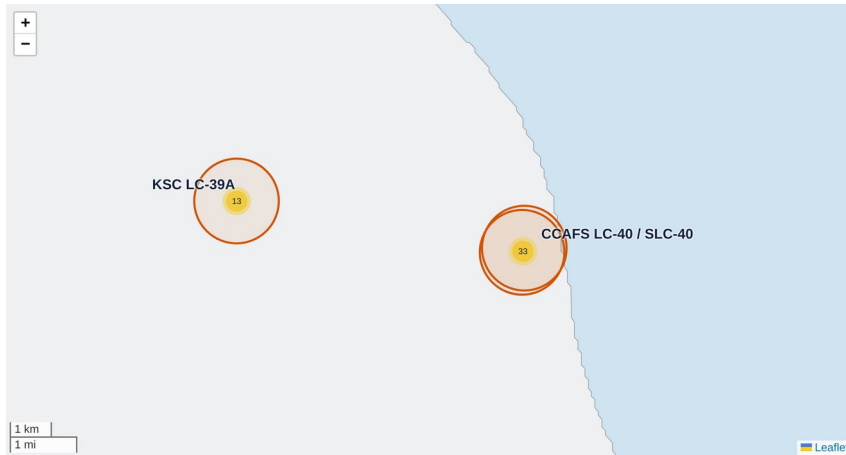


FINDINGS

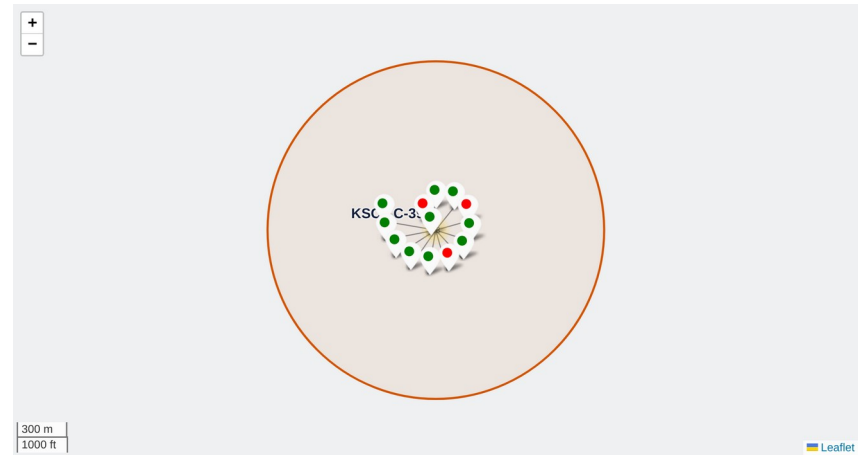
All four sites sit on a coast at the southern edge of the U.S. (close to the Equator). Florida sites launch east over the Atlantic; Vandenberg launches south over the Pacific for polar orbits.

Folium Map — Launch Outcomes per Site

MarkerCluster of every launch: green marker = landed (class 1), red = failed (class 0)



Clusters show 13 launches at KSC LC-39A and 33 at the CCAFS pad



Clicking the KSC LC-39A cluster: 10 green, 3 red markers

FINDINGS KSC LC-39A has the best ratio (10 landed / 3 failed, 76.9%). The CCAFS pad (LC-40 + SLC-40) flew most launches but mostly early ones, so it has the most failures (10 / 23).

Folium Map — Proximity of CCAFS SLC-40

folium.PolyLine from the launch site, distances computed with the haversine formula



Nearest...	Distance
Coastline	0.87 km
Titusville	23.2 km
Cocoa	25.5 km
Melbourne	53.4 km

FINDINGS

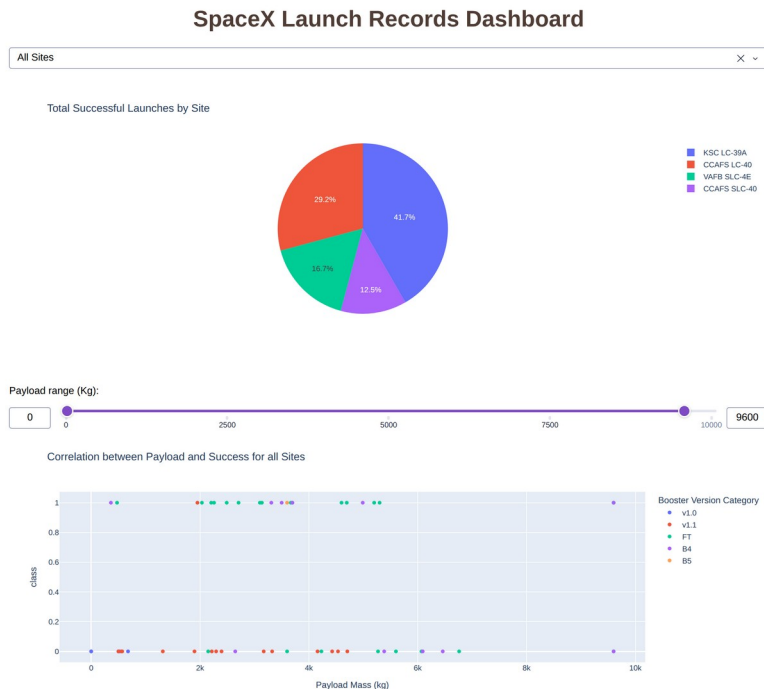
The pad is under 1 km from the sea (safe abort and recovery over water) and 23+ km from the nearest towns for public safety.

04

Plotly Dash Dashboard

Interactive launch-records dashboard (spacex_dash_app.py)

Dashboard Overview



Dropdown (site-dropdown): All Sites or one of the four launch sites.

Pie chart (success-pie-chart): all sites → share of successes per site; one site → success vs. failure.

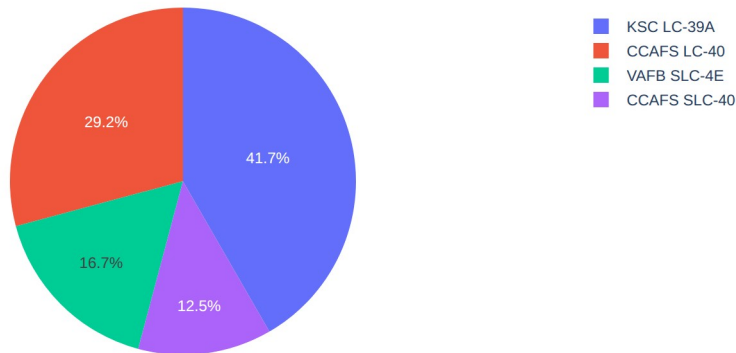
Range slider (payload-slider): filters payload from 0 to 10,000 kg in 1,000 kg steps.

Scatter (success-payload-scatter-chart): payload vs. class, coloured by booster version; reacts to both the dropdown and the slider (two Inputs in one callback).

Dashboard — Successful Launches by Site

Dropdown = All Sites

Total Successful Launches by Site



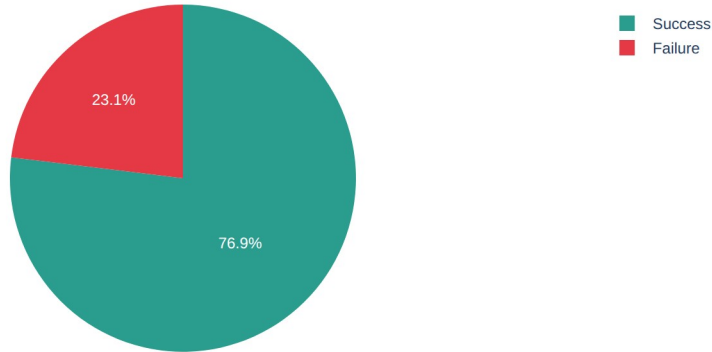
FINDINGS

KSC LC-39A has the largest share of successful launches (41.7%), then CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%) and CCAFS SLC-40 (12.5%). Shares partly reflect how many missions each site flew.

Dashboard — Site with the Highest Success Rate

Dropdown = KSC LC-39A

Success vs. Failed Launches for site KSC LC-39A



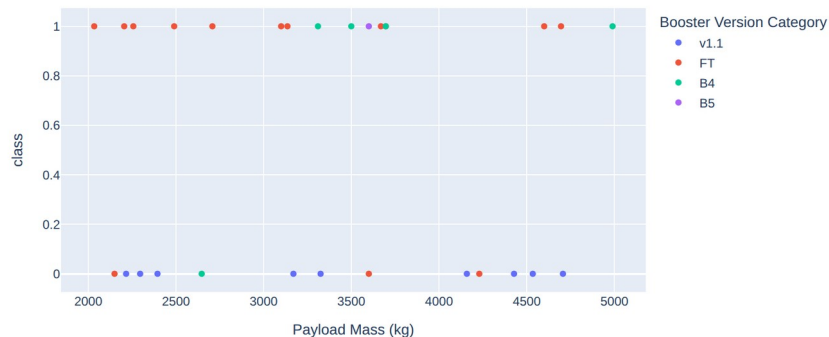
FINDINGS

KSC LC-39A landed 10 of 13 first stages: a 76.9% success rate, the best of any site (CCAFS LC-40 27%, CCAFS SLC-40 43%, VAFB SLC-4E 40%).

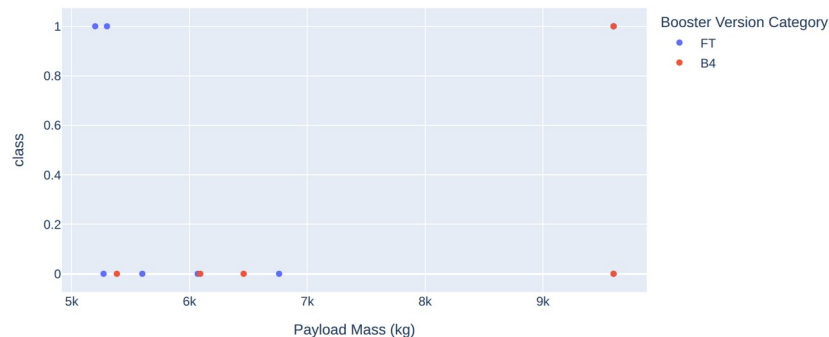
Dashboard — Payload vs. Launch Outcome

Same scatter, two slider settings (all sites)

Correlation between Payload and Success for all Sites



Correlation between Payload and Success for all Sites



Slider 2,000–5,000 kg: 16 of 29 landed (55%)

Slider 5,000–10,000 kg: 5 of 14 landed (36%)

Payload sweet spot: 2,000–5,000 kg has the most successes and the highest rate; below 2,000 kg only 23% landed (mostly early v1.0/v1.1 flights).

Booster matters most: FT boosters landed 16 of 24 (67%), B4 6 of 11; v1.1 landed only 1 of 15 and v1.0 none.

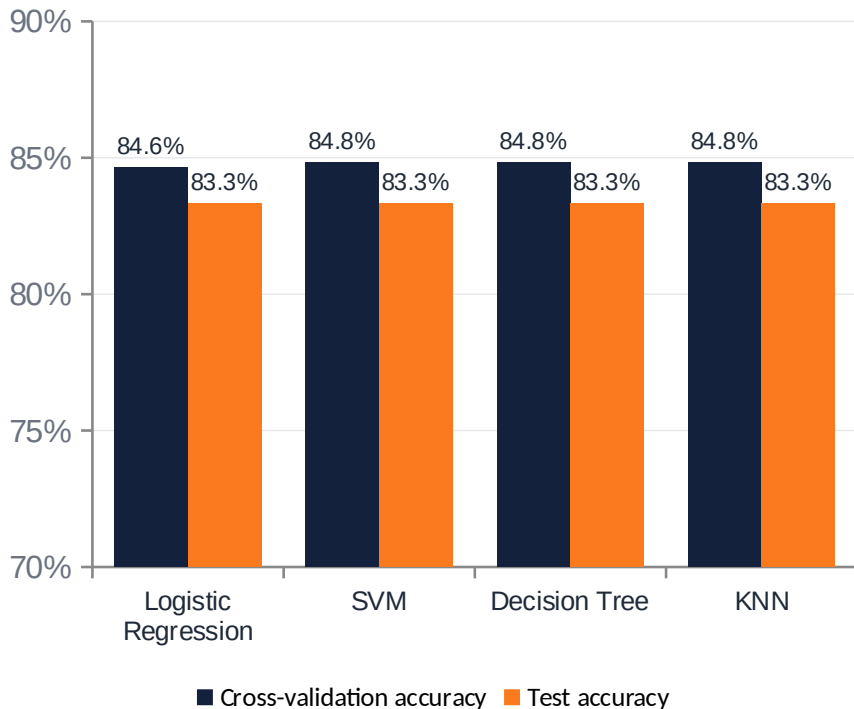
05

Predictive Analysis

Classification results

Classification Accuracy

GridSearchCV (cv=10) on 72 training launches, scored on 18 test launches

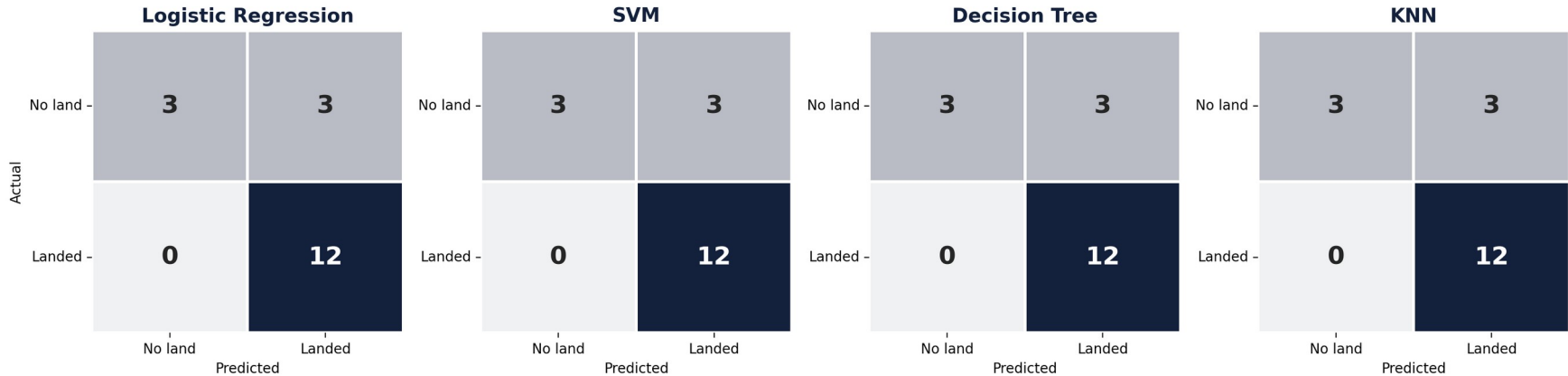


Model	CV	Test	Prec.	Recall	F1
Logistic Regression	84.6%	83.3%	80%	100%	88.9%
SVM	84.8%	83.3%	80%	100%	88.9%
Decision Tree	84.8%	83.3%	80%	100%	88.9%
KNN	84.8%	83.3%	80%	100%	88.9%

FINDINGS

On this split every model gets 15 of 18 right (83.3%), with 100% recall and 80% precision. SVM, decision tree and KNN share the top CV score (84.8%), so one split cannot separate them; see the robustness check on slide 33. Y-axis starts at 70%.

Confusion Matrices — All Four Models

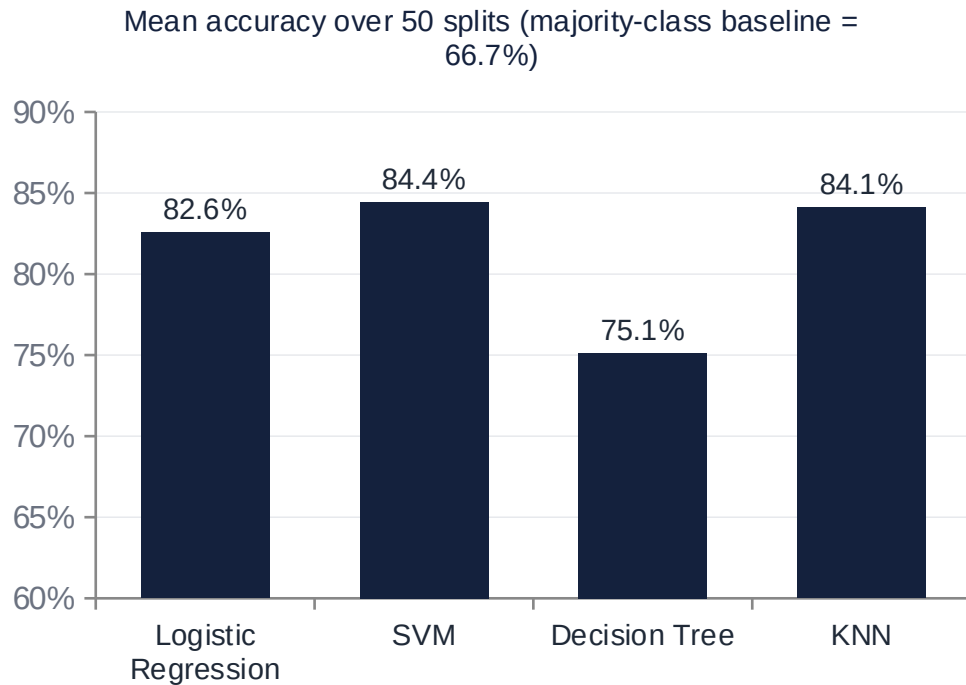


FINDINGS

All four models produce the same matrix: 12 true positives, 3 true negatives, 3 false positives and 0 false negatives. Every landing was caught, but 3 of the 6 failed landings were predicted to land, so the models are over-optimistic about failures.

Which Model Is Best? — Robustness Check

50 different train/test splits (5-fold $\times$ 10 repeats) using each model's tuned hyperparameters



BEST MODEL: SVM

SVM (sigmoid kernel) is the most accurate and stable: $84.4\% \pm 8.0\%$, ahead of KNN (84.1%) and logistic regression (82.6%). The decision tree drops to 75.1%: its single-split tie was luck, and it overfits. All models clearly beat the 66.7% baseline.

Conclusions

1 Experience pays off

Landing success rose from 0% before 2014 to 90% in 2019; flight number is the strongest single signal.

2 Site and orbit matter

KSC LC-39A has the best record (76.9%). GTO is the hardest orbit (52%); SSO, GEO, HEO and ES-L1 landed every time.

3 Hardware and payload

v1.0/v1.1 boosters rarely landed; FT, B4 and B5 account for nearly all successes. 2,000–5,000 kg payloads land most often.

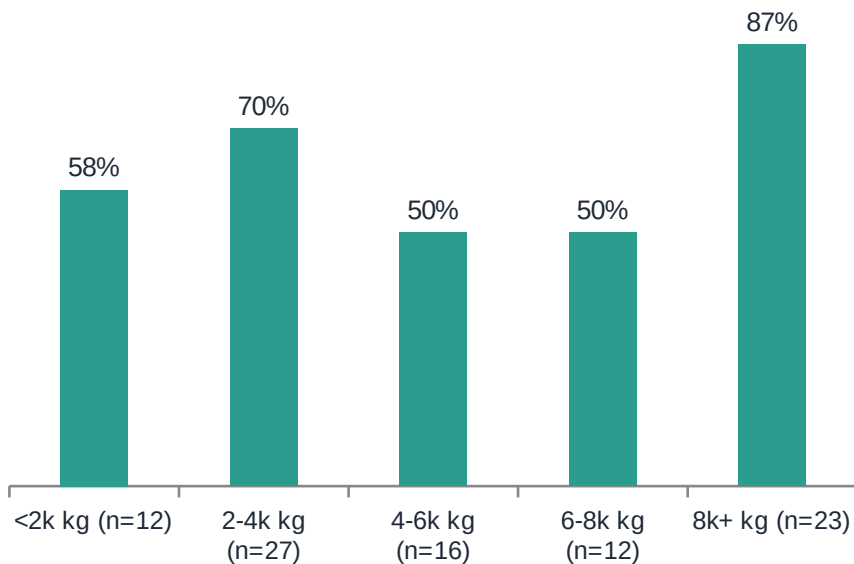
4 SVM is the best model

All models reach 83.3% on the test set; across 50 splits SVM is best (84.4%). Errors are false positives, so treat predicted landings with care.

Innovative Insights — Beyond the Template

What a competing bidder like Space Y should take away

Landing success by payload band



Payload is not a penalty: the heaviest band (8,000 kg+) lands most often (87%), because heavy missions flew late on mature boosters. Era confounds payload.

Price a GTO bid higher: GTO missions land only about half the time, so SpaceX's true cost on those launches is closer to expendable pricing.

One split is not enough: four models tied at 83.3% on the 18-launch test set, but resampling 50 splits showed SVM best and the decision tree overfitting (75%).

Next step: add launch year and booster block as explicit features and retrain on post-2020 launches to test whether accuracy holds.

Appendix

GitHub repository: <https://github.com/kautzarichramsya/Applied-Data-Science-Capstone>

Data sources: SpaceX REST API (IBM static snapshot `API_call_spacex_api.json`), Wikipedia "List of Falcon 9 and Falcon Heavy launches", IBM `Spacex.csv` and `spacex_launch_dash.csv`.

Best hyperparameters:

- Logistic regression: `C=0.01`, `penalty=l2`, `solver=lbgfs`
- SVM (best model): `kernel=sigmoid`, `C=1`, `gamma=0.0316`
- Decision tree: `criterion=gini`, `max_depth=8`, `min_samples_leaf=4`, `splitter=random`
- KNN: `n_neighbors=10`, `p=1`, `algorithm=auto`

Repository contents: notebooks for data collection (API), web scraping, data wrangling, EDA with SQL, EDA with visualization, Folium maps and machine learning prediction, plus `spacex_dash_app.py`.